

[Clo 1-4, SO2] Section 1: Basic Operating System Concepts
[7 Marks] Choose the correct answer. Questions 1-16 each has 0.5 Mark

1) Which one of the following is <u>NOT</u> a process attribute that is stored in PCB?	a) Process name b) memory limits c) process state d) Non-Of the above
2) After a process is terminated,	a) it is moved to terminated state b) its process control block is removed c) if it has children processes, they will be notified d) All of the Above
3) The data segment of a process contains	a) instructions b) Dynamic date c) Return values of methods or functions d) Non-of-the above
4) Operating system designers use to notify the operating system that an external event has been occurred.	a) System calls b) trap c) interrupt d) Non of the above
5) The degree of multiprogramming is	a) the number of processes executed per unit time b) the number of processes in the ready queue c) the number of processes in the I/O queue d) the number of terminated processes
6) Which process can inherit something from other process?	a) cooperating process b) child process c) parent process d) init process
7) In UNIX, the return value for the fork system call is ____ for the child process and ____ for the parent process.	a) A Negative integer, Zero b) Zero, A Negative integer c) Zero, A nonzero integer d) A nonzero integer, Zero
8) A process is terminated if	a) its scheduled quantum is finished b) it receives uncaught kill signal c) if it finished its task d) (a) and (b) e) (b) and (c)
9) The process Control block (PCB) of a process is created at state.	a) new b) ready c) running d) none of the mentioned
10) In a shared memory inter-process communication facility shared area	a) must be of fixed size which is chosen by operating system. b) can be of any size less than hard disk capacity. c) must be of any variable size which is chosen by operating system. d) must be of any variable size which is chosen by the communicating processes.
11) In the message passing with non blocking, the sending process will after the	a) go to wait state b) continue running

send operation.

12) What is true with Timesharing (multitasking) system?

- c) go immediately to ready state.
d) none of the mentioned

- a) Operating system try to utilize the computer hardware resources efficiently.
b) Operating system gives each process the CPU for certain period of time.
c) Application couldn't know the time when they will get running.
d) All-of-the-above.

13) When a computer is first turned on or restarted, a special program calledloads the operating system into memory to be executed.

- a) Compile and Go loader
b) Boot loader
c) Bootstrap loader
d) Relating loader

14) If all processes in a system are I/O bounded processes, then the ready queue will almost be and the Short term Scheduler will have a to do.

- a) full, little
b) full, lot
c) empty, little
d) empty, lot

Answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12	Q13	Q14

[Clo 1-4, SO 2] Section 1: Basic CPU scheduling Concepts

[4 Marks] Choose the correct answer. Questions 16-19 each has 1 Mark

Fill in the following sentences with the correct word. Choose the correct word from the following words: 1) short, 2) hard affinity, 3) Soft affinity, 4) long, 5) non-preemptive 6) preemptive, 7) long

15) means OS may decide to migrate a process to run on another processor.

16) In time sharing system with round Robin CPU scheduler if we choose the quantum too, it will lowers the CPU efficiency but setting the quantum too may cause poor response time to short interactive requests.

17) The CPU Schuler of a batch system is scheduler.

[CLO 1-4, SO 2] Section2: Process Management Concepts.

(16 Marks): Question 18-25 has 2 Mark and Q26 has 6 Marks

18) Mention two general roles of an operating system in today's computer environment.

19) Describe in detail what happened when a process calls a system call. You can use a diagram including the kernel and user mode to show the steps used in executing the system call.

The CPU will switch from user mode to
Kernel Mode $\text{set } \text{mode} = 1$
 $\text{mode} = 0$
2- the kernel executes program
3- the kernel executes code
4- after finish CPU will switch back
to user mode $\text{set } \text{mode} = 1$

20) What is the difference between message passing and shared memory inter-process communication facilities? Make comparison

21) Explain the difference between Kernel and User mode of an operating system?

22) Use a diagram to explain the context switch between 2 processes and describe exactly what happen during switching between them.

Type equation here.

23) Explain by example how and why the context switch could occur during the execution of a process in kernel mode.

24) Mention the role of the operating system short-term scheduler and show you can increase the CPU utilization.

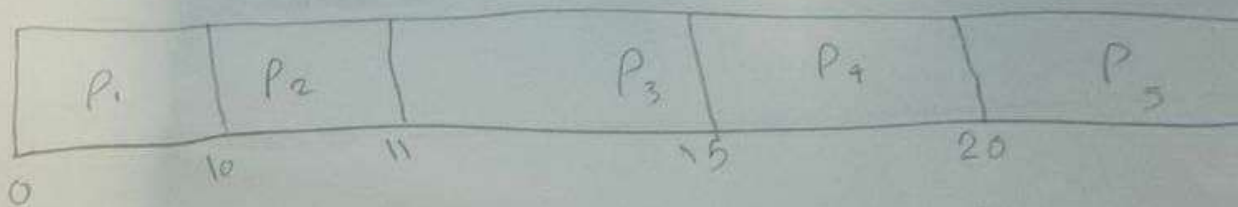
25) What are the criteria of good process scheduler? Mention 4 of them and explain them very briefly.

- 26) For below Processes table, calculate the average waiting time for the following algorithms:
- First Come First Serve (FCFS)
 - NON_Preemptive Shortest Job First (SJF) and
 - Round robin Scheduling Process with 2 nsec quantum.

Draw the Gantt chart and don't ignore the arrival time in your calculations.

Process	Arrival Time	Burst Time
P1	0	10
P2	1	1
P3	2	4
P4	3	5
P5	4	2

FCFS



waiting time =

turnaround

$$P1 = 0$$

$$P2 = 9$$

$$P3 = 9$$

$$10$$

$$10$$